Vliv způsobu balení a doby skladování na fyzikálně chemické, senzorické mikrobiologické parametry dušené vepřové šunky
The influence of the packaging method and time of storage on physicochemical, sensory and microbiological parameters of cooked pork Ham

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ABSTRACT

This study investigated the effect of different packaging methods on the quality of cooked pork ham during refrigerated storage. Samples were evaluated on the day of production and after 10 days of storage at 4 °C. Ham was packaged in paper with polyethylene film, modified atmosphere packaging (MAP), or vacuum packaging. Physico-chemical, microbiological, and sensory parameters were assessed using standardized ISO methods. The results showed that microbiological development depended on the packaging method, while physico-chemical parameters remained within expected ranges for cooked pork Ham.

MATERIAL AND METODOLOGY

Cooked pork Ham was used as a model heat-treated meat product. Samples (250 g) were obtained in 2025, packaged on the day of production, and stored at 4 °C, with analyses performed on Day 1 and Day 10. Physico-chemical, microbiological, and sensory parameters, including selected pathogens, were evaluated according to relevant ISO standards. Sensory analysis was performed by seven trained assessors using unstructured 100-mm scales. Data were log-transformed and analysed by one-way ANOVA with Tukey's test ($p < 0.05$).

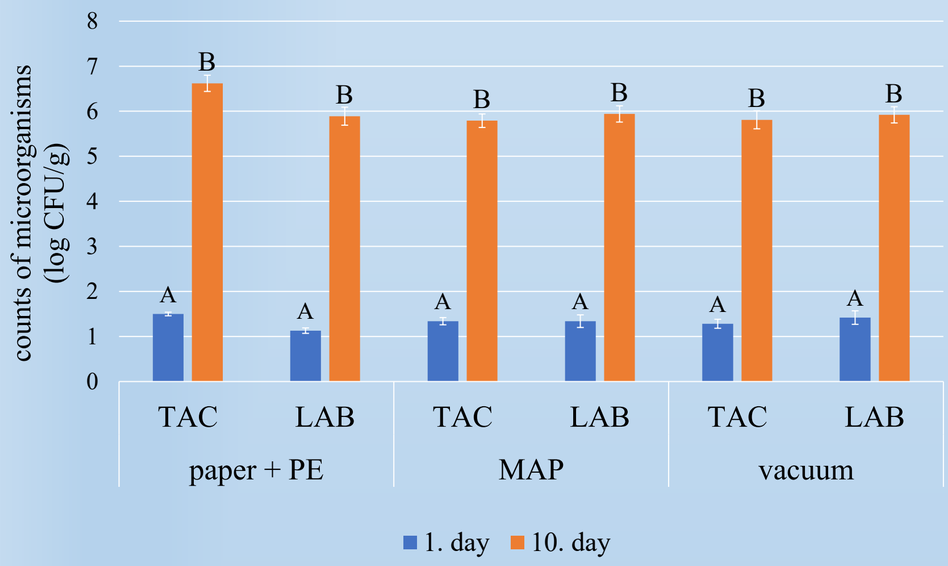

Figure 1. Comparison of the counts total aerobic microorganisms (TAC) and lactic acid bacteria (LAB) respectively in cooked ham the 1st day and the 10th day of stored at 4 °C; the results were calculated respective of the type of the tested packaging (paper + PE, MAP – modified atmosphere and vacuum); A, B – means with different superscripts differ at $P < 0.05$ (according to the storage date within the given packaging); n = 3

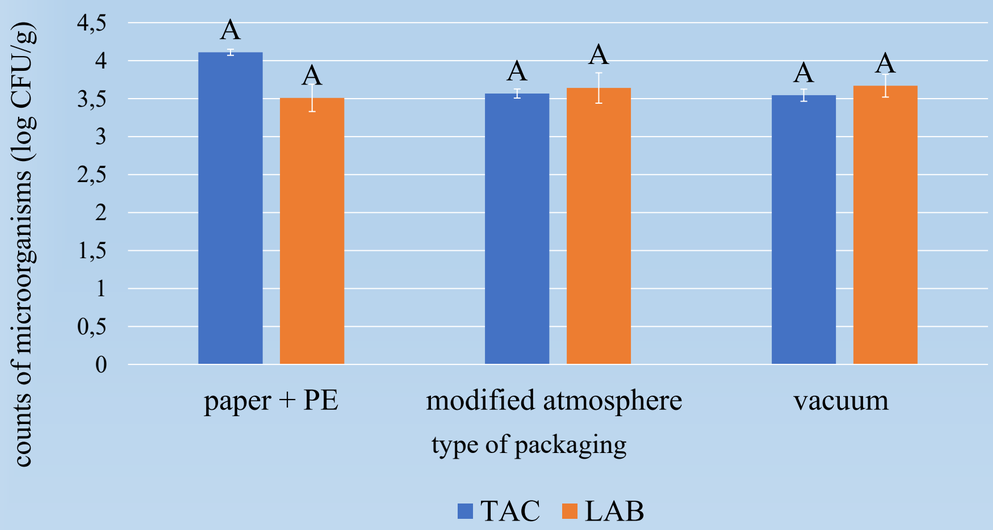

Figure 2 Comparison of the counts total aerobic microorganisms (TAC) and lactic acid bacteria (LAB) respectively in cooked ham stored at 4 °C in different packaging. The results were calculated irrespective of the day of stored, according type of packaking: paper + PE vs. MAP vs. vacuum); n = 6

Packaging	Day	pH	a_w	NaCl (%)
Paper+PE	1	$5.8 \pm 0.00^{a,A}$	$0.97 \pm 0.001^{a,A}$	$1.89 \pm 0.001^{a,A}$
Paper+PE	10	$5.8 \pm 0.002^{a,A}$	$0.97 \pm 0.002^{a,A}$	$1.89 \pm 0.001^{a,A}$
MAP	1	$5.8 \pm 0.00^{a,A}$	$0.97 \pm 0.00^{a,A}$	$1.89 \pm 0.00^{a,A}$
MAP	10	$5.8 \pm 0.001^{a,A}$	$0.97 \pm 0.001^{a,A}$	$1.89 \pm 0.001^{a,A}$
Vacuum	1	$5.8 \pm 0.00^{a,A}$	$0.97 \pm 0.00^{a,A}$	$1.89 \pm 0.00^{a,A}$
Vacuum	10	$5.8 \pm 0.001^{a,A}$	$0.97 \pm 0.001^{a,A}$	$1.89 \pm 0.001^{a,A}$

Table 1 Physicochemical characteristics of cooked ham. a – means no statistické differences in sensory parameters between the 1st and 10th day of storage within a given package. A – means no statistical difference in sensory parameters between different packaging methods (paper + PE vs. MAP vs. vacuum) regardless of storage time; n = 6

Packaging	Day	Appearance & colour	Texture	Aroma	Overall acceptability
Paper+PE	1	$95 \pm 7^{a,A}$	$92 \pm 12^{a,A}$	$94 \pm 9^{a,A}$	$93 \pm 12^{a,A}$
Paper+PE	10	$60 \pm 12^{b,A}$	$55 \pm 10^{b,A}$	$50 \pm 9^{b,A}$	$52 \pm 12^{b,A}$
MAP	1	$94 \pm 12^{a,A}$	$92 \pm 11^{a,A}$	$94 \pm 10^{a,B}$	$93 \pm 7^{a,B}$
MAP	10	$88 \pm 10^{b,A}$	$85 \pm 12^{b,A}$	$90 \pm 10^{b,B}$	$87 \pm 8^{b,B}$
Vacuum	1	$94 \pm 9^{a,A}$	$92 \pm 11^{a,A}$	$94 \pm 12^{a,B}$	$93 \pm 10^{a,B}$
Vacuum	10	$87 \pm 10^{b,A}$	$86 \pm 10^{b,A}$	$89 \pm 11^{b,B}$	$88 \pm 9^{b,B}$

Table 2: Sensory evaluation of cooked ham (stored at 4 °C) by packaging type and storage day. Means marked with different letters are statistically different ($P < 0.05$), a, b – mean statistical differences in sensory parameters between the 1st and 10th day of storage within a given package. A, B – letters mean statistical differences in sensory parameters between different packaging methods (paper + PE vs. MAP vs. vacuum) regardless of the storage time of the ham; n = 6

RESULTS AND DISCUSSION

Cooked pork Ham was evaluated on Day 1 and Day 10 of refrigerated storage (4 °C) using three packaging systems: Paper+PE, MAP, and vacuum.

Physicochemical properties: Initial pH (5.7–5.8), water activity (~0.97), and NaCl content (1.89 g/100 g) corresponded to typical cooked ham. After 10 days, only minor changes were observed, with a slight pH increase in Paper+PE, while aw and NaCl remained stable.

Microbiological evaluation: Pathogens were not detected at any storage time. Indicator microorganisms were low on Day 1 and increased by Day 10 (TAC 5.81–6.22 log CFU/g; LAB 5.89–5.94 log CFU/g), reflecting normal microbial development. No significant differences between packaging types were observed ($P > 0.05$).

Sensory assessment: Day 1 samples showed high acceptability with uniform colour, good texture, and clean aroma. After 10 days, slight declines in appearance and juiciness occurred, mainly in Paper+PE, while MAP and vacuum maintained higher sensory quality. Appearance and texture were the most sensitive quality indicators.

CONCLUSION

Cooked pork Ham, characterized by low salt content, pH near 6, and high water activity, is susceptible to microbial growth. Physicochemical and microbiological parameters met regulatory and quality standards on Day 1 and Day 10 of refrigerated storage. While physicochemical properties remained stable, microbial development depended on packaging type, with MAP and vacuum providing better control than simple packaging. These findings highlight the importance of packaging selection for maintaining product safety and quality.

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